Metro Flux — Australia (Sydney Metro / Transport for NSW)

Access Basics

Everything comes from one platform: the **Transport for NSW (TfNSW) Open Data Hub**, opendata.transport.nsw.gov.au.

1. Register at the Open Data Hub, create an "Application" to get your **API key**.
2. Send it as a header on every request: `Authorization: apikey YOUR_KEY` (not a URL parameter).
3. Free tier ("Bronze Plan"): 60,000 calls/day, 5 req/sec — plenty for this project.
4. Licence: CC BY 4.0 — credit Transport for NSW as the source.

This is a priority-only list — the datasets worth manually extracting for Metro Flux, in the order to pull them. Everything else researched (taxi ranks, non-metro modes, historical occupancy, GTFS-Pathways, etc.) is skipped as low-value for this project.

Priority 1: GTFS-Realtime Vehicle Positions v2 — live train locations

This is Sydney's standout dataset — live GPS positions, something neither London nor Tokyo Metro has.

Base path: `https://api.transport.nsw.gov.au/v2/gtfs/vehiclepos/`

Endpoints (call the one you need — `metro` is the priority): - `GET /metro` — Sydney Metro live vehicle positions - `GET /sydneytrains` — Sydney Trains live vehicle positions - `GET /lightrail/innerwest` — Inner West Light Rail

Response is Protocol Buffers binary by default. Add `?debug=true` to get readable text instead — confirmed working:

```
GET https://api.transport.nsw.gov.au/v2/gtfs/vehiclepos/metro?debug=true
Header: Authorization: apikey YOUR_KEY
```

Returns a `FeedMessage` with a `header` (timestamp) and repeated `entity` blocks, each containing a `vehicle` with `trip_id`, position (lat/long), and timestamp — this is your live train-tracking signal.

Priority 2: Timetables Complete GTFS (static) — foundation dataset

Link: Timetables Complete GTFS

Standard static GTFS zip. Pull these files from it: - `stops.txt` — every station/stop, name, ID, lat/long (your station list) - `routes.txt` / `trips.txt` — line definitions and how stations connect (your network graph) - `stop_times.txt` — scheduled arrival/departure baseline to compare live running against - `shapes.txt` — precise line-route geometry for map drawing

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Priority 3: GTFS-Realtime Trip Updates v2 — delays & cancellations

Link: Public Transport – Realtime Trip Update v2

Base path: `https://api.transport.nsw.gov.au/v2/gtfs/realtime/` (check exact sub-path in the dataset's Swagger spec — same pattern as Vehicle Positions v2: separate endpoints per mode, likely `/sydneytrains`, `/metro`, `/lightrail/innerwest`). Same auth and `?debug=true` trick apply.

Trip-level delay, cancellation, changed-stopping-pattern, and replacement-vehicle data — more granular than a line-status flag.

Priority 4: GTFS-Realtime Alerts v2 — disruptions

Link: Public Transport – Realtime Alerts v2

Stop/trip/line-level disruption and planned-works alerts, covering Train, Metro, Bus, Ferry, Light Rail. Equivalent to TfL's Disruptions API. Same auth pattern.

Priority 5 (optional): Trip Planner API

Link: Trip Planner APIs

Origin → destination journey options with live adjustments and Opal fares built in. Nice-to-have for alternative-route suggestions, not essential to the core load-balancing logic.

Skipped (researched, not worth pulling)

- **Historical Train Occupancy (Nov 2018–Feb 2019)** — outdated, not live, low value now that you're prioritizing Vehicle Positions as your real-time signal.
- **GTFS-Pathways** — accessibility extension, not core to load-balancing.
- **Non-v2 Realtime feeds (Vehicle Positions / Trip Update without "v2")** — broader (all NSW modes incl. buses/ferries/regional), unnecessary noise vs. the v2 feeds scoped to Trains/Metro/Light Rail.
- **Taxi Ranks, Stop Finder, Departure Board, Metro/Network map PDFs** — nice reference material, not data your system needs to ingest.

What You Won't Get

- No CCTV access.
- No live per-carriage crowding API — historical dataset only (skipped above), so treat crowding as an estimate/proxy problem for Sydney, same as Tokyo Metro.
- No internal operational data (signalling, staffing).

Bottom line: Vehicle Positions v2 is your headline real-time signal for Sydney — something Tube doesn't have at all. Pair it with static GTFS for structure and Trip Updates/Alerts v2 for disruption context.